Initial Model Running — Preliminary Results

Model Status

> The system is operational. The honeypot accepts live SSH connections, the LLM-backed terminal responds in real time, MITRE tags fire automatically, and the dashboard displays all sessions live.

Current working components confirmed running:

- SSH honeypot server (honeypot.py) — accepts connections on port 2222
- LLM terminal simulator (llm_engine.py) — GPT-4o-mini via OpenRouter
- Real-time MITRE ATT&CK tagger (state_engine.py) — 20 technique rules
- Live analyst dashboard (dashboard.py) — Streamlit, auto-refreshes every 2s
- Prompt evolution engine (prompt_evolution.py) — rewrites rules every 5 commands
- Narrative engine (narrative_engine.py) — generates coherent company world per session
- Red team agent (red_team.py) — evaluates detection risk every 10 commands

Preliminary Results*

*all of the testing was done in a controlled environment by simulated agents

The following results were collected from synthetic attacker sessions run against the live system. Four attacker profiles were simulated — each running a scripted command sequence characteristic of that attacker type.

Session Summary Table

Attacker Profile	Commands	Dwell Time	MITRE Techs	Canary Hit	Privilege
Script Kiddie	14	38s	6	Yes	root
Intermediate	23	3m 42s	9	Yes	root
APT	31	11m 20s	12	Yes	root
Botnet	9	12s	3	No	user

MITRE ATT&CK Techniques Observed

Across all sessions, the following MITRE ATT&CK techniques were detected:

- T1033 — System Owner Discovery (whoami, id, who)
- T1003 — Credential Dumping (cat /etc/passwd, cat /etc/shadow)
- T1548 — Privilege Escalation (sudo -l, sudo su, CVE exploit)
- T1046 — Network Service Scanning (nmap 10.0.0.0/24)
- T1552 — Unsecured Credentials (cat ~/.ssh/id_rsa, cat config.php)
- T1053 — Scheduled Task/Job (crontab -l, cat /etc/crontab)
- T1105 — Ingress Tool Transfer (wget, curl to external IPs)
- T1082 — System Information Discovery (uname -a, cat /etc/os-release)
- T1057 — Process Discovery (ps aux, top)
- T1005 — Data from Local System (cat /opt/app/config.json)

Key Findings*

*all of the testing was done in a controlled environment by simulated agents

Finding	Detail
World coherence	Employee names generated for /etc/passwd appear in auth.log and bash histories — cross-agent consistency confirmed
Canary effectiveness	3 out of 4 attacker profiles accessed planted credential files (config.php, config.json) within 5 minutes

Finding	Detail
Prompt evolution	After 5 commands, system correctly identified credential_hunting focus in 2/4 sessions and injected relevant bait files
Dialogue engine	APT profile triggered mail command, dialogue engine generated realistic 3-turn sysadmin conversation
Kill chain depth	APT profile reached Installation stage (5/7 kill chain stages) — deepest engagement observed
Response latency	Cached commands (ls, whoami, id) respond in <50ms. Novel LLM commands average 380ms via OpenRouter